

ENGINEERING MEMO

To:	Plant Project Review Board
From:	Process Engineering Review
Date:	28 June 2026
Subject:	CH ₄ / C ₂ H ₄ splitter shortcut FUG validation
Revision:	A

Modelling basis

The validation treats the service as a binary methane/ethene split at 7.0 bar using the provided constant relative volatility $\alpha(\text{CH}_4/\text{C}_2\text{H}_4) = 1.85$. The feed is subcooled with $q = 1.2$. The kettle reboiler is included as an equilibrium stage; the total condenser is excluded from the stage count. Results are shortcut estimates and do not include hydraulic capacity, tray/packing efficiency, or rigorous thermodynamic regression.

Overhead-GC interpretation

The analyser is specified as a vapor-line GC above the total condenser, so the reported CH₄:C₂H₄ vapor peak ratio of 18.0:1.00 was normalized directly as the distillate composition. This gives $x_{D,\text{CH}_4} = 0.9474$ and $x_{D,\text{C}_2\text{H}_4} = 0.0526$; the bottoms specification is $x_{B,\text{CH}_4} = 0.0100$ and $x_{B,\text{C}_2\text{H}_4} = 0.9900$.

Shortcut results and balance check

Result	Value	Comment
Underwood root theta	1.1243	Root selected between $\alpha_{\text{HK}} = 1.000$ and $\alpha_{\text{LK}} = 1.85$
Minimum reflux ratio R_{min}	0.9918	Back-substitution gives $R_{\text{min}} + 1 = 1.9918$
Operating reflux ratio R	1.2397	Specified as $1.25 \times R_{\text{min}}$
N_{min}	12.17	Includes kettle reboiler; excludes total condenser
Gilliland/Eduljee N	27.29	Continuous theoretical stage estimate at operating reflux
D/F and B/F	0.7793 / 0.2207	Molar basis; $D/F + B/F = 1.0000$

Engineering limitations for board review

The shortcut package supports a consistent review basis, but final design lock should still check sensitivity to α and q , confirm the GC/distillate interpretation, and compare the continuous theoretical stage requirement against the actual column stage count, efficiency basis, pressure profile, and hydraulic constraints.